Primitive Root

Assignment:

1. Show that 2 is primitive root modulo 11

A number g is a primitive root modulo p (prime p) if its order modulo p is $\varphi(p) = p-1$. For $p = 11$ we need the order of 2 modulo 11 to be 10.

Compute powers of 2 mod 11:

$2^1 \equiv 2$ $\quad 2^6 \equiv 20 \equiv 9$

$2^2 \equiv 4$ $\quad 2^7 \equiv 18 \equiv 7$

$2^3 \equiv 8$ $\quad 2^8 \equiv 14 \equiv 3$

$2^4 \equiv 16$ $\quad 2^9 \equiv 6$

$2^5 \equiv 10$ $\quad 2^{10} \equiv 12 \equiv 1 \pmod{11}$

Thus the order of 2 modulo 11 is 10, so, 2 is primitive root modulo 11.

2. How many incongruent primitive roots does 14 have ?

Ans: If primitive roots exists for n, their number is $\varphi(\varphi(n))$. Primitive roots exist for $n = 2, 4, p^k, 2p^k$ where p is an odd prime. Here $14 = 2 \cdot 7 = 2p$ with $p = 7$, so primitive roots do exist.

$$\varphi(14) = \varphi(2)\varphi(7) = 1 \cdot 6 = 6$$
$$\varphi(\varphi(14)) = \varphi(6) = 2$$

So, there are 2 incongruent primitive roots modulo 14.

3. a) show $ord_n a = ord_n(a^{-1})$

Let $ord_n(a) = d$. By definition $a^d \equiv 1 \pmod n$ and d is the least positive integer with their property.

$$(a^{-1})^d = a^{-d} \equiv 1 \pmod n$$

So, the order of a^{-1} divides d: $ord_n(a^{-1}) \mid d$

Conversely, if $\text{ord}_n(a^{-1}) = e$ then

$$a^e = ((a^{-1})^e)^{-1} \equiv 1^{-1} \equiv 1 \pmod{n}$$

So, $d \mid e$. Combining the two divisibilit

$\text{ord}_n(a^{-1}) \mid d$ and $d \mid \text{ord}_n(a^{-1})$ gives

$\text{ord}_n(a^{-1}) = d = \text{ord}_n(a)$.

b) If a is a primitive root modulo n then $\text{ord}_n(a) = \varphi(n)$. For

Ans: From part (a) we have $\text{ord}_n(a^{-1}) = \text{ord}_n(a)$.

Hence, $\text{ord}_n(a^{-1}) = \varphi(n)$, so a^{-1} is also a primitive root modulo n.